IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-DS-004

1. Topic (12 words max)

Data Driven Tax Intelligence: Predictive and Advisory System for Tax Management in Sri Lanka

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Data Science - DS

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sri Lanka's personal and corporate tax management process is complex and constantly changing due to frequent policy updates and the technical language used in regulations issued by the Inland Revenue Department of Sri Lanka [1]. As a result, many individuals, small and medium enterprises (SMEs), accountants, and financial advisors struggle to correctly understand and apply tax rules. Manual interpretation of tax laws often leads to calculation errors, non-compliance, and financial penalties, which is a common issue in developing tax systems [2]. The main users of the proposed system are salaried employees, business owners, and tax practitioners who require accurate, timely, and personalised tax guidance.

The impact of this problem is wide-ranging. Socially, low tax awareness increases stress and reduces voluntary compliance among taxpayers. Economically, inefficient tax planning causes unnecessary financial losses for individuals and limits effective revenue collection for the government [4]. From a technical perspective, existing tax-related systems lack automation, predictive capabilities, and clear explanations. Ethically and legally, incorrect tax calculations can result in unintentional violations, disputes, and penalties. In addition, continued reliance on paper-based documentation and manual processes contributes to inefficiency and resource wastage [2].

This research falls under the Faculty of Computing (FoC) Research Cluster in Technology Integration and Management and aligns with the FinTech and Government Digital Transformation industry sectors. By applying data-driven techniques such as rule-based systems, predictive analytics, optimisation methods, and explainable artificial intelligence, the project aims to address the limitations of current tax management approaches and support digital tax reform initiatives [4].

Furthermore, the research problem supports several United Nations Sustainable Development Goals (SDGs). It contributes to SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), and SDG 16 (Peace, Justice and Strong Institutions) by improving transparency, trust, and compliance in public tax administration [3].

References

1. Inland Revenue Department of Sri Lanka (2023) *Inland Revenue Act No. 24 of 2017 and amendments*. Colombo: IRD Sri Lanka. Available at: https://www.ird.gov.lk (Accessed: 7 January 2026).
2. Organisation for Economic Co-operation and Development (2020) *Tax Administration 3.0: The Digital Transformation of Tax Administration*. Paris: OECD Publishing. Available at: https://www.oecd.org/tax/administration/ (Accessed: 7 January 2026).
3. United Nations (2015) *Transforming our World: The 2030 Agenda for Sustainable Development*. New York: United Nations. Available at: https://sdgs.un.org/goals (Accessed: 7 January 2026).
4. World Bank (2021) *Digital Government and Tax Administration Reform*. Washington, DC: World Bank. Available at: https://www.worldbank.org (Accessed: 7 January 2026).

6. Brief description of Existing Research and Systems (300 words max)

Several products and systems currently exist to support tax calculation and compliance. In Sri Lanka, the Inland Revenue Department of Sri Lanka provides online platforms such as RAMIS, which allow taxpayers to register, submit tax returns, and make tax payments electronically [1]. These systems improve administrative efficiency and reduce paperwork. However, they mainly function as submission and record-keeping platforms and do not actively assist users in understanding or calculating their tax liabilities. Taxpayers or accountants are required to manually calculate tax amounts based on applicable tax laws before entering values into the system, which often leads to errors caused by misinterpretation of complex regulations [2].

In addition to government platforms, some private tools such as the SimpleBooks online income tax calculator provide basic tax estimations using predefined tax slabs. While such tools are useful for quick reference, they are unofficial, rely on static tax rules, and offer limited personalisation or legal assurance [2]. As a result, these systems are insufficient for comprehensive tax planning or decision support.

Existing research in tax and financial technology has explored the use of rule-based expert systems, data analytics, and machine learning techniques to improve tax compliance and financial decision-making. Several studies apply natural language processing (NLP) to analyse legal and financial documents, while predictive models are used to estimate tax liabilities and detect anomalies [3]. Optimisation techniques have also been proposed to support better financial planning by identifying efficient allocation of tax reliefs [4]. More recent research highlights the importance of explainable artificial intelligence (XAI) in financial systems, as it improves transparency, accountability, and user trust [5].

However, much of the existing research focuses on isolated components such as prediction, fraud detection, or compliance monitoring rather than developing complete end-to-end tax decision-support systems. Furthermore, many studies are evaluated using datasets from developed countries, which may not reflect the regulatory complexity, frequent policy changes, and institutional constraints present in developing economies such as Sri Lanka [2], [6].

Despite these advances, several gaps remain. Most current systems do not automatically interpret changes in tax laws or convert legal text into machine-readable rules. They also lack intelligent tax optimisation, future tax prediction, and policy impact analysis, which are essential in volatile tax environments [1], [4]. In addition, limited explainability makes tax outcomes difficult for users to understand and trust. There is no integrated solution that combines legal rule automation, predictive modelling, optimisation, and clear explanations within a single framework tailored to the Sri Lankan context. These gaps highlight the need for a data-driven and explainable tax intelligence system, which this research aims to develop.

References - Existing Research and Systems

1. Inland Revenue Department of Sri Lanka, Inland Revenue Act No. 24 of 2017 and Amendments. Colombo: IRD Sri Lanka, 2023. Available: https://www.ird.gov.lk
2. Organisation for Economic Co-operation and Development, Tax Administration 3.0: The Digital Transformation of Tax Administration. Paris: OECD Publishing, 2020.
3. K. Sharma and R. Kumar, "Application of machine learning techniques in tax compliance systems," Expert Systems with Applications, vol. 115, pp. 255–267, 2019.
4. T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794, 2016.
5. M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you? Explaining the predictions of any classifier," Proceedings of the 22nd ACM SIGKDD Conference, pp. 1135–1144, 2016.
6. World Bank, Digital Government and Tax Administration Reform. Washington, DC: World Bank, 2021.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Overview

The proposed solution is an intelligent, end-to-end Personal Income Tax (PIT) decision-support system tailored to the Sri Lankan tax context. It transforms tax compliance from a manual, error-prone process into a transparent, data-driven, and proactive one through the integration of legal-rule automation, optimization algorithms, predictive analytics, and explainable AI. This solution continuously adjusts to IRD policy changes, recommends the most efficient tax-saving methods,projects future tax burdens under uncertainty, and provides human-readable explanations foreach decision, in contrast to current systems that rely on static rule updates and basic calculators.

Addressing Identified Research Gaps

This research directly addresses key gaps identified in existing tax systems:

The primary weaknesses of the current personal income tax systems in Sri Lanka are addressed in this study. The majority of existing systems are slow to adjust to frequent legislative changes since they rely on static, manually updated regulations. In order to ensure ongoing legal compliance, this study mechanically extracts and converts tax laws from official Inland Revenue Department papers into machine-readable representations.

Current methods don't offer strategic tax planning and are mainly restricted to basic tax computation.z
Furthermore, the majority of tools only offer historical or current insights. Forward-looking tax estimates that take policy uncertainties and economic trends into consideration are presented in this study. Lastly, thorough validation and concise, human-readable explanations enhance transparency and boost user compliance and trust.

Finally, because taxpayers frequently find it difficult to figure out how results are obtained, many systems lack transparency and user trust. By offering concise, comprehensible explanations for tax results and suggestions, this study fills this knowledge gap.

Component-wise System Perspective (Input – Process – Output)

Component 1: Legal Rules Automation & Core Calculation Engine

- **Input:** IRD tax acts, circulars, policy updates, user income details, expenses, donations, and bank statements
- **Process:** Extract tax rules using NLP, convert legal text into JSON/YAML rules, process financial documents using OCR, and compute baseline tax liability via a core calculation engine
- **Output:** Machine-readable tax rules, categorised financial data, accurate baseline tax payable, and a reusable Tax Calculation API

Component 2: Optimization & Strategy Engine

- **Input:** Baseline tax values, structured tax rules, eligible reliefs, and user preferences (risk, effort, liquidity)
- **Process:** Apply constraint-based optimisation to identify the best combination of tax reliefs and rank strategies using multi-criteria analysis
- **Output:** Optimised tax-saving strategies and ranked recommendations for legal tax reduction

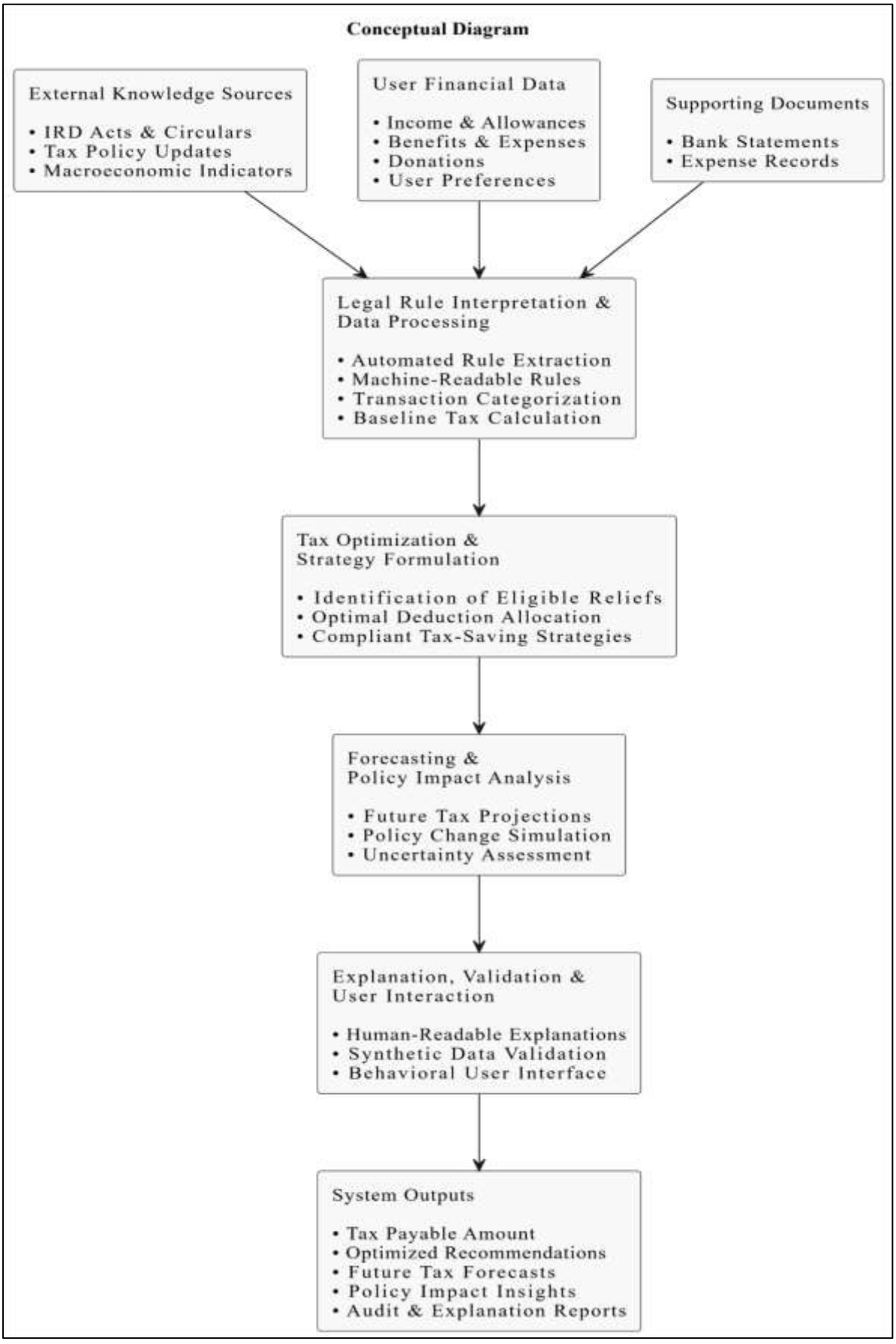
Component 3: Predictive & Policy Module

- **Input:** Historical income data, economic indicators, tax rules, and potential policy changes
- **Process:** Use hybrid time-series models to forecast future tax liabilities and simulate policy impact scenarios
- **Output:** Future tax projections with uncertainty ranges and personalised policy impact insights

Component 4: Explanatory Interface & Validation Module

- **Input:** Outputs from all previous components and synthetic validation datasets
- **Process:** Generate human-readable explanations using XAI techniques and validate results through large-scale automated testing
- **Output:** Clear explanations for tax outcomes, validation reports, and improved user trust and compliance

Conceptual Diagram



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful execution of the proposed intelligent tax decision-support system depends on the availability of specialized domain expertise, technical knowledge, and reliable data sources. This project is feasible within an academic research setting, as the required expertise and data can be systematically acquired through structured learning, publicly available resources, and controlled data generation methods.

Gaining Expertise in a Particular Domain

Accurately modeling personal income tax regulations and compliance requirements requires specialized domain expertise in public finance and taxation. By carefully reviewing the publicly accessible official Inland Revenue Department (IRD) Acts, amendments, circulars, and interpretation guidelines, team members can acquire this knowledge. Furthermore, domain understanding will be enhanced by referencing secondary materials such tax handbooks, expert opinions, and academic research papers on tax systems and fiscal policy. Further validation of interpretations and assurance of legal accuracy can be achieved by consulting with accounting experts, tax specialists, or university supervisors with taxation experience. This methodical approach guarantees that subject knowledge is based on reliable and authentic sources

Gaining Technical Knowledge and Proficiency

Multidisciplinary knowledge in data engineering, natural language processing, optimization methods, predictive analytics, and explainable systems are needed for this project. Through a combination of coursework, online learning environments, and practical experimentation, team members can develop these skills. Well-established libraries and frameworks that are supported by extensive documentation and research material can be used to develop technical competencies like NLP for legal text processing, rule-based system design, constraint programming, time-series forecasting, and explainable AI. While peer participation and supervisor comments will assist assure methodological rigor and correctness, prototyping and iterative implementation will further strengthen practical understanding.

Data Collection and Availability

Sensitive or confidential datasets are not necessary to obtain the data needed for this project. Official government publications provide public access to legal and policy information, such as tax legislation, circulars and updates. In order to replicate true taxpayer scenarios , financial input data including bank statements, income records, and expense details, will be created artificially. Without affecting individual privacy, synthetic data generation allows controlled experimentation, scalability and through system validation.

Additionally, publicly accessible information released by international organizations and government agencies can provide macroeconomic indicators useful for forecasting, such as trends in income growth or inflation rates.

Ethical Considerations and Clearance Requirements

Access to actual private or sensitive taxpayer data is not needed for this project. For development, testing and the assessment, all user financial records will be either fake or artificially created. Consequently, no human subjects, personal identifiers or sensitive personal data are used in the study. As a result, formal ethical clearance is not necessary. However, by guaranteeing data privacy, being transparent abut model behavior, and using prediction outputs responsibly, ethical best practices will be followed. Future additions will seek the proper ethical permission in compliance with institutional rules if they contain actual user data or user studies

9. Objectives and Novelty

Main Objective To develop an intelligent, end-to-end Personal Income Tax (PIT) decision-support system for Sri Lanka that automates legal rule interpretation, optimizes tax-saving strategies, forecasts future tax liabilities under policy uncertainty, and provides human-readable, validated explanations to improve taxpayer compliance and trust.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22064486	Design and develop a **Legal Rules Automation and Core Calculation Engine** to interpret IRD policy changes and process raw financial data into machine-readable formats.	• Build an NLP model to extract key tax entities from IRD circulars/PDFs. • Implement Rule-as-Code to convert legal text into JSON/YAML rule-sets. • Develop an OCR-based Bank Statement Processor to categorize transactions into deductible buckets. • Create the foundational Tax Calculation API.	Moves away from static, manual updates by introducing **automated rule extraction** from legal PDFs. It converts complex legal prose directly into executable logic, ensuring the system is always compliant with the latest IRD amendments.
IT22118936	Develop an **Intelligent Recommendation and Strategy Engine** that mathematically optimizes the allocation of tax reliefs to minimize liability.	• Apply Constraint Programming (CP) to model tax laws as mathematical constraints. • Build a Multi-Criteria Recommendation system to rank strategies by risk, effort, and liquidity. • Formulate logic for income restructuring into exempt non-cash benefits.	Unlike basic calculators, this uses **Constraint Programming** to find the absolute mathematical optimum for tax savings. It provides a **Formal Mathematical Proof of optimality**, which is rare in standard commercial tax tools.

IT22896186	Create a **Predictive and Policy Module** to forecast future tax burdens and simulate the personal impact of potential government policy changes.	• Build a Hybrid Time-Series model (ARIMA + LSTM) to predict salary growth and inflation. • Develop a Policy Volatility Quantification tool to provide confidence intervals for tax projections. • Create a "What-If" Policy Simulator for testing hypothetical tax-free limit changes.	Introduces the concept of a "**Tax Time Machine**." It quantifies **Policy Uncertainty** as a metric, allowing users to plan for Sri Lanka's volatile tax landscape by seeing a range of future liabilities rather than just a single number.
IT22238580	Design an **Explanatory Interface and Automated Validation Framework** to provide transparent reasoning and prove system accuracy through rigorous auditing.	• Apply XAI (LIME/SHAP) to rule-based systems to generate human-readable "Why" explanations. • Build a Behavioral Nudging UI using economics principles to encourage compliance. • Create a Monte Carlo Simulation framework for large-scale synthetic data validation.	Applies **Explainable AI (XAI)** specifically to a **rule-based engine**, translating logic-gate results into plain English. It also uses **Monte Carlo simulations** to generate thousands of test cases, providing academic rigor and a formal error report.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				